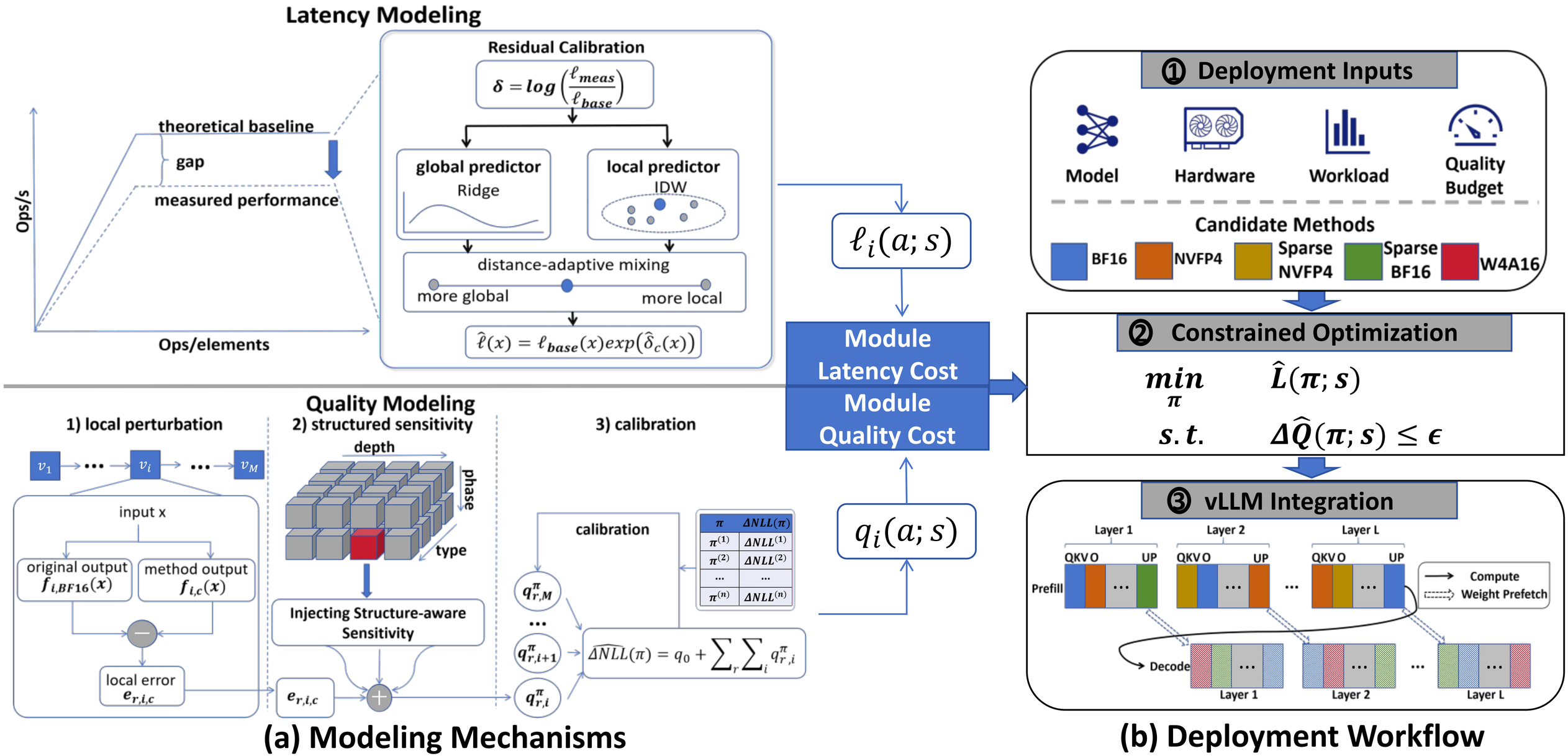

Latency Modeling
theoretical baseline
gap
measured performance
Ops/s
Ops/elements
Residual Calibration
$\delta = log\left(\frac{\ell_{meas}}{\ell_{base}}\right)$
global predictor
Ridge
local predictor
IDW
distance-adaptive mixing
more global
more local
$\hat{\ell}(x) = \ell_{base}(x)exp(\hat{\delta}_c(x))$
$\ell_i(a;s)$
Module Latency Cost
Module Quality Cost
$q_i(a;s)$
Quality Modeling
1) local perturbation
2) structured sensitivity
3) calibration
v_1
v_i
v_M
input x
original output $f_{i,BF16}(x)$
method output $f_{i,c}(x)$
local error $e_{r,i,c}$
depth
phase
type
Injecting Structure-aware Sensitivity
$e_{r,i,c}$
calibration
π | $\Delta NLL(\pi)$
$\pi^{(1)}$ | $\Delta NLL^{(1)}$
$\pi^{(2)}$ | $\Delta NLL^{(2)}$
... | ...
$\pi^{(n)}$ | $\Delta NLL^{(n)}$
$q^{\pi}_{r,M}$
$q^{\pi}_{r,i+1}$
$q^{\pi}_{r,i}$
$\widehat{\Delta NLL}(\pi) = q_0 + \sum_r \sum_i q^{\pi}_{r,i}$
① Deployment Inputs
Model
Hardware
Workload
Quality Budget
Candidate Methods
BF16
NVFP4
Sparse NVFP4
Sparse BF16
W4A16
② Constrained Optimization
$\min_{\pi} \quad \hat{L}(\pi; s)$
$s.t. \quad \Delta\hat{Q}(\pi; s) \le \epsilon$
③ vLLM Integration
Layer 1
Layer 2
Layer L
QKV
O
UP
Prefill
Compute
Weight Prefetch
Decode
(a) Modeling Mechanisms
(b) Deployment Workflow